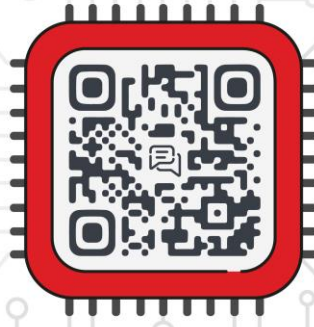




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نظري

كلية الهندسة المعلوماتية

السنة الثانية

Signals

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RB Informatics; 06/04/2026

الاتصالات الرقمية

السلام عليكم ورحمة الله وبركاته

Course contents:

- Basic Definitions of digital communications.
- Signal classification factors

Definitions:

- **Digital:** Digital refers to electronic technology that generates, stores and processes data using binary code-a series of 0s and 1s (bits).
- **Communication:** is the process of exchanging information, ideas, thoughts and meaning between individuals or groups.

In network communication, the communicating entities are applications.

Example: When a WhatsApp message is sent, the communication is not exactly between two people or two mobile phones; it is between the WhatsApp application on the sender side and the WhatsApp application on the receiver side.

- **Digital communication:** is the transfer of information, data, or messages using digital signals, typically encoded in binary (0s and 1s) and transmitted over electronic networks, including internet, wireless or storage channels.
- **communication system:** is a collection of components including a transmitter, channel and receiver that work together to exchange information, such as voice, data, or video, from a source to a destination.



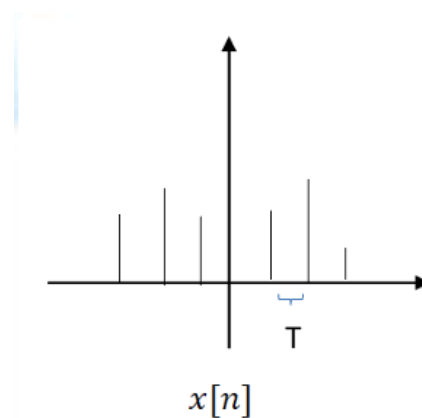
كيف نحصل على $x[n]$ من $x(t)$ ؟
نأخذ samples ...

Sampling interval: time between 2 samples

$$\begin{aligned} & x(t_0), x(t_1), \dots, x(t_n) \\ \text{or } & x[0], x[1], \dots, x[n] \\ \text{So: } & \text{or } x_0, x_1, \dots, x_n \\ & x_n = x[n] = x[t_n] \end{aligned}$$

A discrete-time signal can be obtained from a continuous-time signal by taking samples every fixed sampling period T .

$x[n]$ is a sequence of samples or numbers, where n is an integer sample index.



Signal classification based on Amplitude:

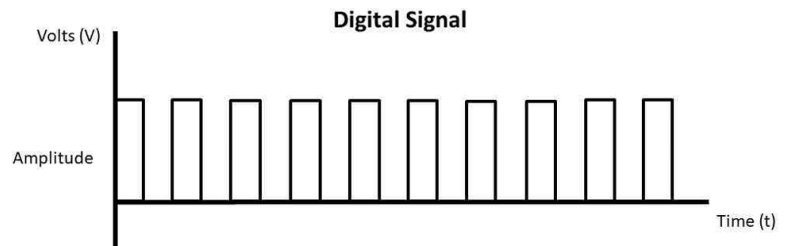
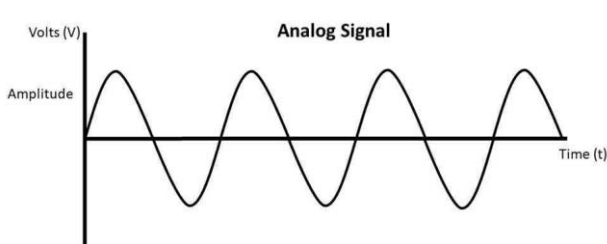
Analog signal: is a continuous, time-varying signal that represents physical measurements like: Sound, Light, Temperature, or Pressure.

Digital Signal: is a discontinuous, time-discrete signal that represents data using a sequence of discrete values typically binary code (0s and 1s) representing high / low voltage levels.

Addition from RBCs:

A digital signal is a special case of a discrete signal whose values are usually limited to two levels: 0 and 1.

An analog signal can be considered a continuous-time signal studied over a specific range.



Examples:

Analog Signal: A microphone for recording sound: When you speak, the microphone converts your voice waves into an electrical voltage. This voltage rises and falls smoothly and with very precise values to match your tone of voice.

Digital Signal: Data transmission over an Ethernet cable: The data does not produce smooth waves, but rather very fast electrical pulses that tell the computer: "There is electricity (1)" or "There is no electricity (0)."

Homework (1):

Why digital signals are preferred in communications?



Signal classification based on periodicity:

- **Periodic signals** repeat a specific pattern over a set period.

(where T is the period and $f = 1/T$ is the frequency.)

Example:

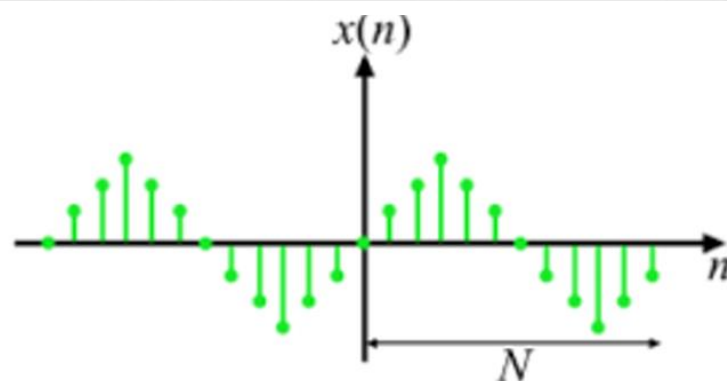
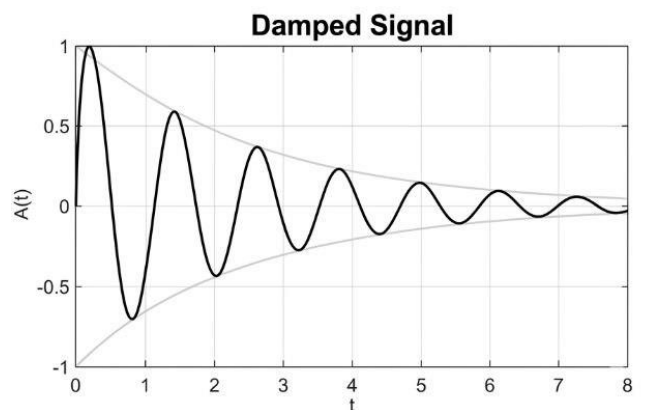
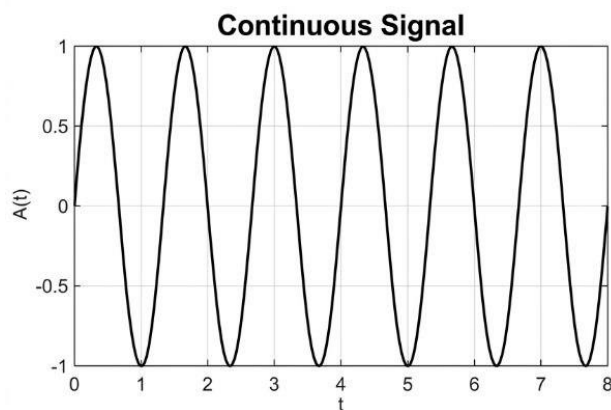
AC power voltage: the 50/60 sine wave from a wall outlet or human heartbeats can also be considered periodic signals.

- **Aperiodic signal:** (non-periodic) signals change constantly without a repeating cycle

Example:

Human speech: uttered words and sentences are unique.

Examples of signal applications include AM/FM radio, short waves, digital signal processing, digital filters, and geological monitoring using sensors.



$$x[n] = x[n + N]$$

For discrete-time periodic signals: $x[n] = x[n + N]$, where N is the period measured in samples.

Signal classification based on symmetry:

Even signals are symmetric about the vertical axis ($X(t) = X(-t)$), creating a mirror image.

Example:

$$X(t) = |t|, \text{DC} \quad X(10) = X(-10)$$

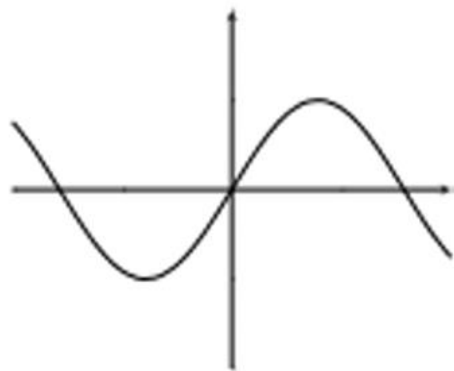
Odd signals are antisymmetric about the origin ($X(t) = -X(-t)$)

Example: $\sin(t)$

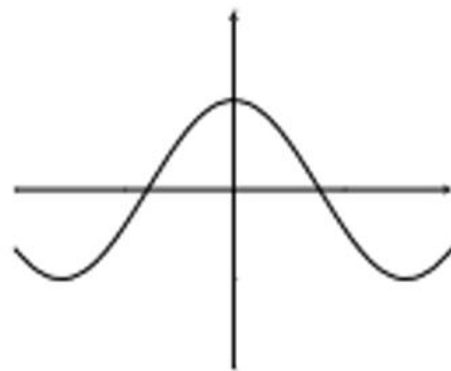
For discrete-time signals:

even signals satisfy $x[n] = x[-n]$,

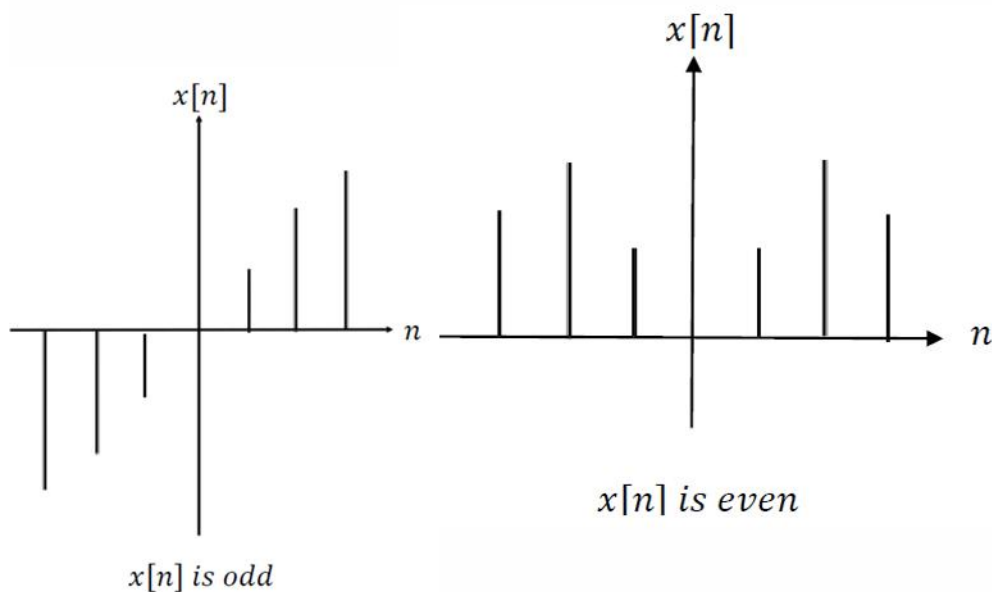
while odd signals satisfy $x[n] = -x[-n]$.



$\sin x$ (odd)



$\cos x$ (even)



Signal classification based on predictability:

Deterministic signals are predictable and can be described by a mathematical formula

Example:

$X(t) = \sin(t)$. $\sin$'s range is always between -1 and +1 or Dc signal

Random signals are unpredictable and probabilistic; their exact values cannot be determined in advance.

Example:

Speech signals, jamming

Signal classification based on energy & power:

Energy signal: A signal is said to be an energy signal if and only if its total energy E is **finite**, i.e. $0 < E < \infty$ for an energy signal, the average power $p=0$

Example:

Nonperiodic signals

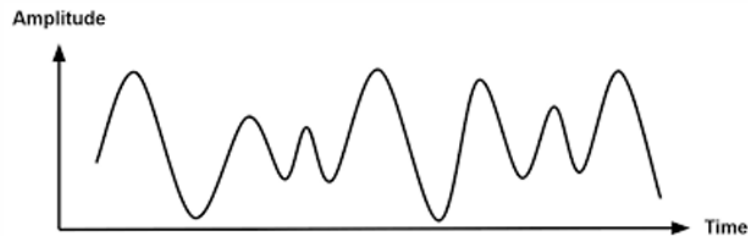
Power signal: A signal is called a power signal if its average power P is **finite**, i.e. $0 < P < \infty$.

For a power signal, the total energy $E = \infty$

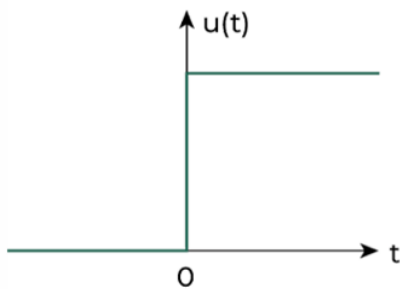
Example: periodic signals

<u>Power signals</u>	<u>Energy signals</u>
1- Normalized average power is finite and non zero ($0 < p < \infty$).	Normalize average energy is finite and non zero ($0 < E < \infty$)
2- Power of CT signal : $p = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)^2 dt$	Energy of CT signal : $E = \int_{-\infty}^{+\infty} x(t) ^2 dt$
3- Power of DT signal : $p = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n) ^2$	Energy of DT signal : $E = \sum_{n=-\infty}^{\infty} x(n) ^2$
4- Practical periodic signals are power signals.	Non periodic signals are energy signals.

Basic CT signals

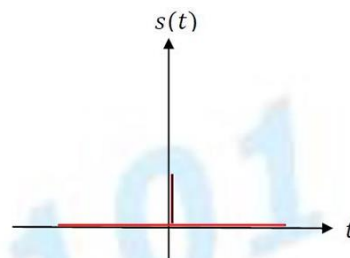


Unit – step function
Heaviside unit function



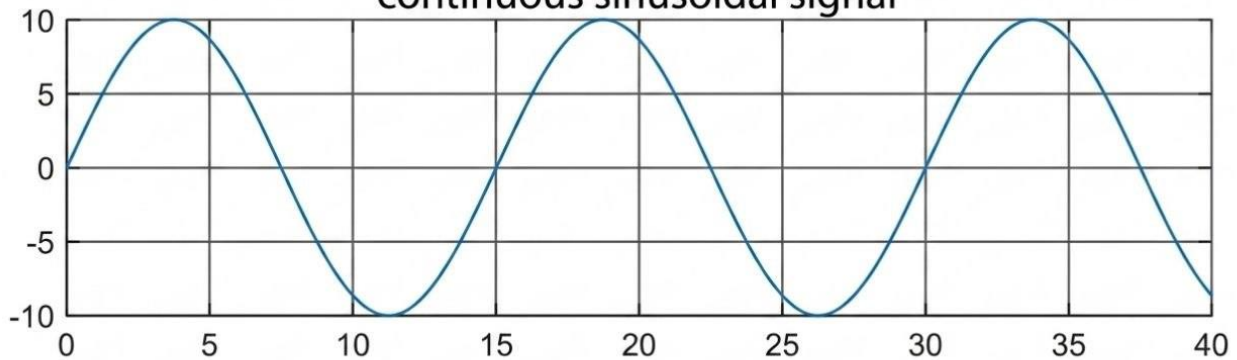
$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Unit impulse function
(Dirac delta function)

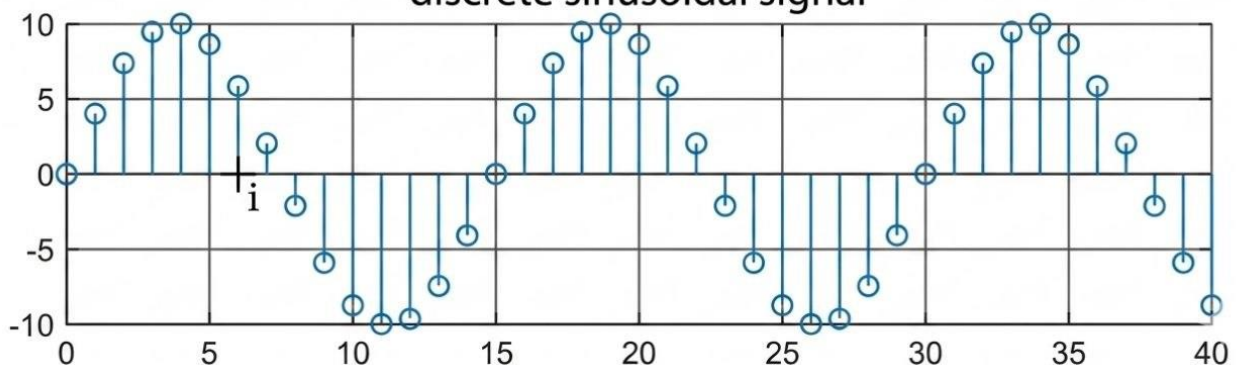


$$s(t) = \begin{cases} 0 & t \neq 0 \\ 1 & t = 0 \end{cases}$$

continuous sinusoidal signal



discrete sinusoidal signal



ملاحظة:

توضح الصورة الفرق بين الإشارة الجيبية المستمرة (في الأعلى) والإشارة المنفصلة (في الأسفل)، حيث يتم تمثيل الإشارة المنفصلة كمجموعة من العينات المأخوذة عند فترات زمنية محددة.

Sinusoidal signal

$$x(t) = A \cdot \cos(\omega_0 t + \varphi)$$

A is the amplitude, which represents the maximum value of the sinusoidal signal.

W: radian frequency.

φ : Phase angle (radian)

$$T_0 = \frac{2\pi}{\omega_0} \quad , \quad f_0 = \frac{1}{T_0} \quad , \quad \omega_0 = 2\pi f_0$$

Benefit: A sinusoidal signal can be represented using either cosine or sine:

$$x(t) = A \cos(\omega_0 t + \varphi) \text{ or } x(t) = A \sin(\omega_0 t + \varphi).$$

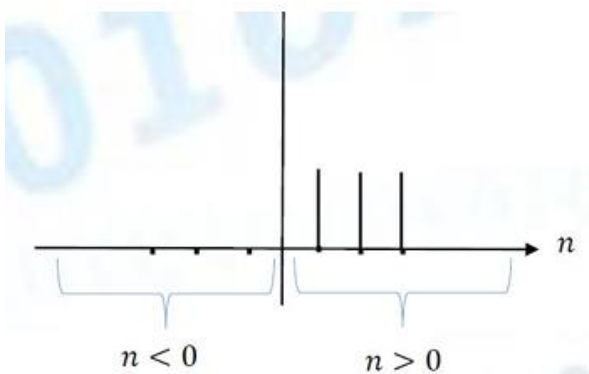
Homework (2):

Sinusoidal signal with different phase values: 0, 90, 180.

Basic discrete - time signals:

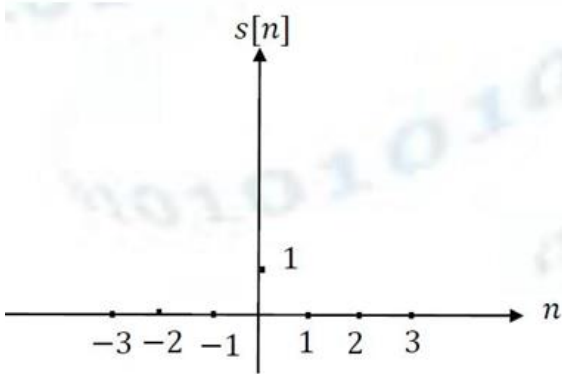
1- Unit-step sequence:

$$u[n] = \begin{cases} 1 & ; \quad n \geq 0 \\ 0 & ; \quad n < 0 \end{cases}$$



2- Unit-impulse sequence:

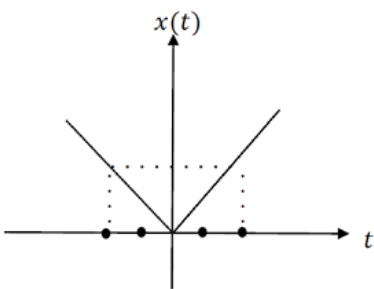
$s[n] = \begin{cases} 1 & ; \quad n = 0 \\ 0 & ; \quad n \neq 0 \end{cases}$



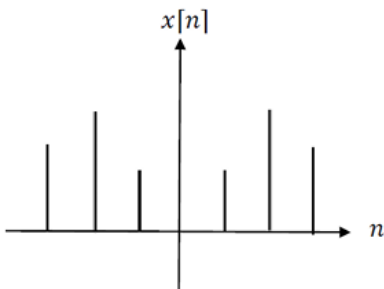
Note: we can replace (s) with (Σ)

x(t) is even if : $x(-t) = x(t)$

x[n] is even if : $x[-n] = x[n]$

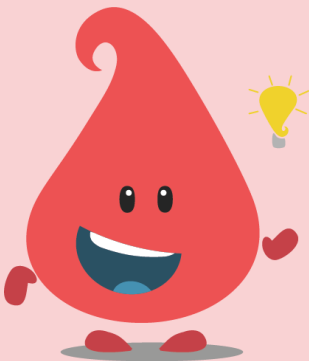


x(t) is even



x[n] is even

RBCs



In the world of sampling, don't take negative samples during moments of weakness... Take samples at the peak of your concentration